STA2002: Probability and Statistics II
Tests of Statistical Hypotheses–Part 1

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Introduction

- In this lecture we will introduce the basic notions and concepts in hypothesis testing.
- In particular, we would discuss hypothesis testing for mean.
- **Suggested reading**: Chapter 8.1 of the textbook.

Hypothesis Testing Framework

to solve decision problems.

We will first introduce the hypothesis testing framework and define some important notions:

- H_0 and H_1 $\Rightarrow$ $H_0 \cap H_1$ contains all the possibilities.
- Test statistic T and critical region C
- Type I and Type II error
- Significance level α
- p-value

Formulate H_0 and H_1

- Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim}$ from a common distribution with unknown parameter μ.
- We have an initial belief/claim/assumption on the values that μ can take, and we call this the **null hypothesis** H_0.
- We would like to test this belief/claim/assumption against a competing belief/claim/assumption, and we call this **alternative hypothesis** H_1.
- A **simple hypothesis** is a hypothesis in which μ only take on one value, e.g., $\mu = 5$.
- A **composite hypothesis** is a hypothesis in which μ takes on a range of values, e.g., $\mu > 5$.

Formulate H_0 and H_1

Examples of H_0 and H_1: ($H_0 \cup H_1$) represents all possibilities in the decision.

- $H_0 : \mu = 5, \quad H_1 : \mu = 6.$
- (One-sided hypothesis test) $H_0 : \mu = 5, \quad H_1 : \mu > 5.$
- (One-sided hypothesis test) $H_0 : \mu = 5, \quad H_1 : \mu < 5.$
- (Two-sided hypothesis test) $H_0 : \mu = 5, \quad H_1 : \mu \neq 5.$
- $H_0 : \mu > 5, \quad H_1 : \mu \leq 5.$

sample space of sample $X_1, \ldots X_n$

data is the realization of sample

Test Statistic T and Critical Region C

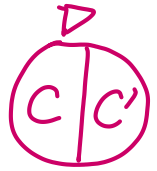
- Once we formulate H_0 and H_1, we collect data, say $x_1, \ldots, x_n$, to see whether H_0 is favoured or H_1 is favoured.
- Then we need to come up with a rule or region, based on the data $x_1, \ldots, x_n$, to determine whether H_0 is rejected or we fail to reject H_0.
- The region that we reject H_0 is called **critical region** C, and the region is usually specified in terms of **test statistic** T.

a decision rule

Decision Rule and Critical Region C

- Denote the space of the sample by $\mathcal{D}$, that is,

$$\mathcal{D} := \text{space}\{(X_1, X_2, \ldots, X_n)\}.$$



$C \cap C' = \emptyset$
$C \cup C' = \mathcal{D}$

- A test of H_0 versus H_1 is based on a subset C of $\mathcal{D}$. This set C is called the **critical region** and its corresponding decision rule (test) is,

↳ a subset of sample space
a subset of $\mathbb{R}^n$.

$$\text{Reject } H_0 \text{ (Accept } H_1\text{), if } (X_1, \ldots, X_n) \in C,$$
$$\text{Retain } H_0 \text{ (Reject } H_1\text{), if } (X_1, \ldots, X_n) \notin C.$$

X_1: random sample
x_1: observation.

Test Statistic T and Critical Region C

Example 1

Example 8.1-1 in book: Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, 36)$. We want to test

$$H_0 : \mu = 50, \quad H_1 : \mu = 55.$$

Solution. We can take the critical region $C = \{\overline{x} \geq 53\}$ and test statistic $\overline{X} \sim N(\mu, 36/n)$. That is, if $\overline{x} \geq 53$, we reject H_0. Otherwise, if $\overline{x} < 53$, we fail to reject H_0. □

Errors in Hypothesis Testing: Type I and Type II Error

- We may draw incorrect conclusion and commit errors in hypothesis testing. It can happen in two ways:

	H_0 true	H_1 true
Reject H_0	**Type I error**	Correct ☺
Fail to Reject H_0	Correct ☺	**Type II Error**

Errors in Hypothesis Testing: Type I and Type II Error

Define the probabilities as follows,

a probability

$$\alpha = P(\text{Test makes Type I Error}) = P(T \in C; H_0).$$

$$\beta = P(\text{Test makes Type II Error}) = P(T \notin C; H_1).$$

Example 2

Example 8.1-1 in book continued: Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, 36)$. We want to test

$$H_0 : \mu = 50, \quad H_1 : \mu = 55.$$

Errors in Hypothesis Testing: Type I and Type II Error

Solution. We can take the critical region $C = \{\overline{x} \geq 53\}$ and test statistic $\overline{X} \sim N(\mu, 36/n)$. Suppose that $n = 16$. Under H_0, $\overline{X} \sim N(50, 36/16)$, and under H_1, $\overline{X} \sim N(55, 36/16)$. As a result,

$$\alpha = P(\overline{X} \geq 53; H_0) = P\left(Z \geq \frac{53 - 50}{\sqrt{36/16}}\right) = 0.0228.$$

$$\beta = P(\overline{X} < 53; H_1) = P\left(Z < \frac{53 - 55}{\sqrt{36/16}}\right) = 0.0913.$$

standard normal

$\mu = 55$

$$P\left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} < \frac{53-\mu}{\sigma/\sqrt{n}}; \mu = 55\right)$$

□

Note, by changing the critical region C, α increases (decreases) while β decreases (increases), so both errors cannot be reduced at the same time.

Errors in Hypothesis Testing: Type I and Type II Error

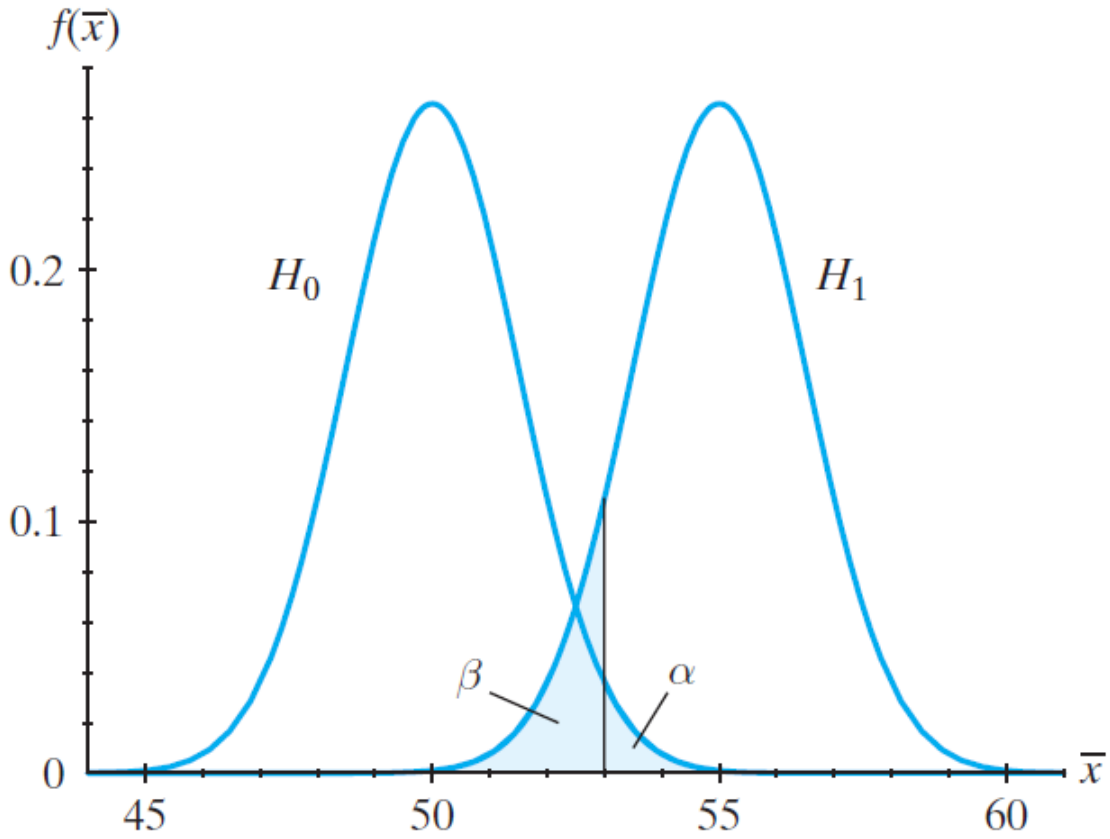


Figure 8.1-1 pdf of $\overline{X}$ under H_0 and H_1

Significance Level

- Most of the time we fix the probability of type I error α, for example, we take $\alpha = 0.05 = 5\%$, and determine the corresponding critical region.
- The value of α is called the **significance level** of the test.
- We can also be more conservative, and for example, take $\alpha = 0.01$.

let α smaller
(set critical region larger) $\Rightarrow$ easier to reject H_0

p-value: Observed Significance Level

- p-value is the probability of observing a more extreme test statistic (in the direction that favours H_1), given that H_0 is true.
- A small p-value indicates the data is unlikely coming from H_0, and so the data is perhaps better explained by H_1. We reject H_0 if the p-value $\leq \alpha$, where α is the significance level.
- p-value is also called the observed significance level.

Hypothesis Testing for Normal Mean

Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$. We would like to conduct hypothesis testing on μ, and the discussion is separated into the following two cases:

- Case 1: σ^2 is known
- Case 2: σ^2 is unknown

Case 1: σ^2 is known

- We consider the test statistic

$$T := \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

- Under $H_0 : \mu = \mu_0$, $T \sim N(0,1)$.
- Once we observe the data $x_1, \ldots, x_n$, we can compute

$$\boxed{t = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}}}.$$

Case 1: σ^2 is known:
(One-sided test) $H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$

- p-value is

$$p = P(T > t; H_0) = P(Z > t),$$

so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = P(Z > t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \geq z_\alpha}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin [\overline{x} - z_\alpha \frac{\sigma}{\sqrt{n}}, \infty)}}_{\text{confidence interval approach}}$$

Case 1: σ^2 is known:
(One-sided test) $H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$

- p-value is

$$p = P(T < t; H_0) = P(Z < t),$$

so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = P(Z < t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \leq -z_\alpha}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin \left(-\infty, \overline{x} + z_\alpha \frac{\sigma}{\sqrt{n}}\right]}}_{\text{confidence interval approach}}$$

Case 1: σ^2 is known:
(Two-sided test) $H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$

- p-value is

$$p = P(|T| > |t|; H_0) = P(Z > |t|) + P(Z < -|t|) = 2P(Z > |t|),$$

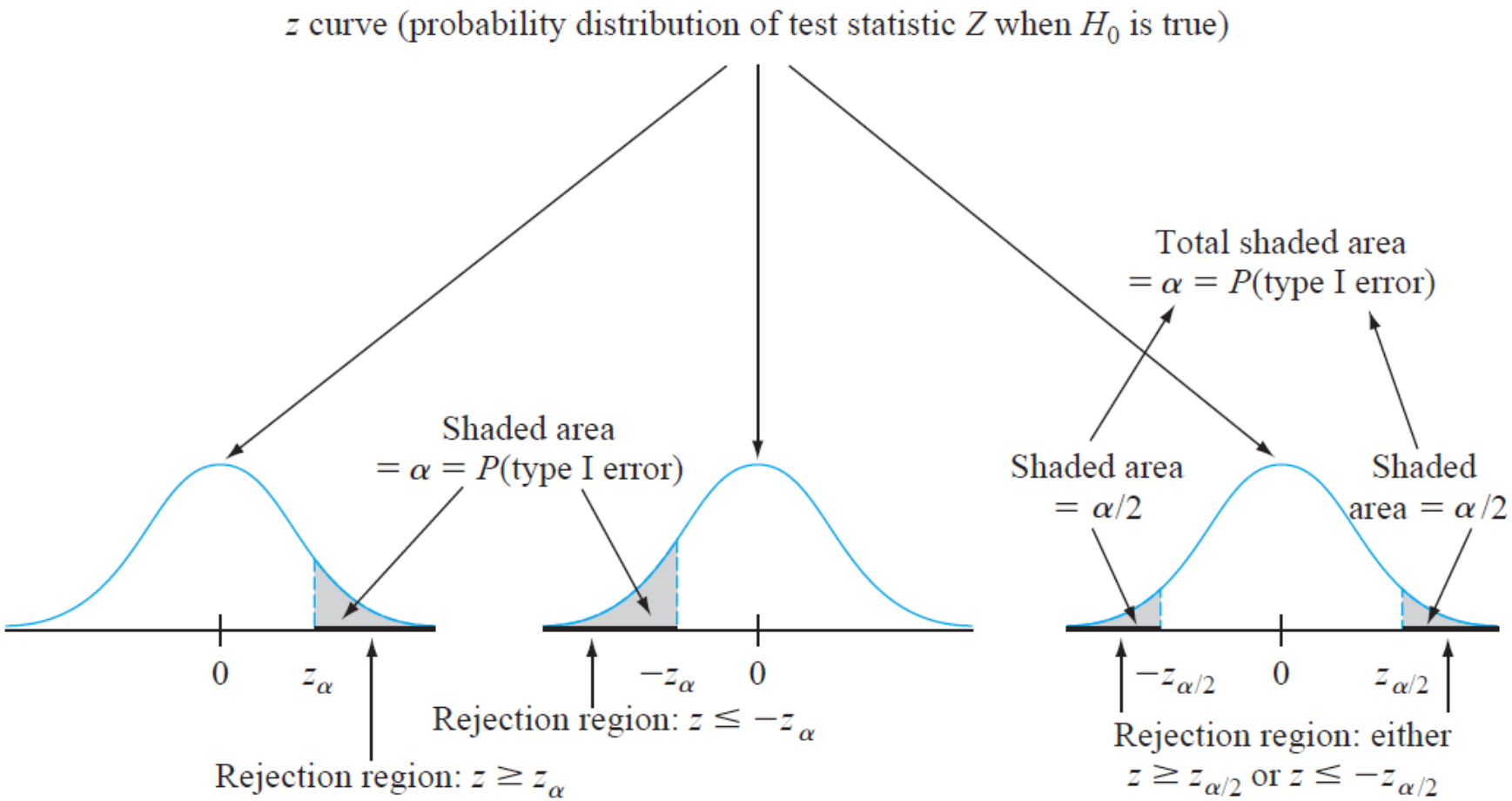
so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = 2P(Z > |t|) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{|t| \geq z_{\alpha/2}}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin [\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]}}_{\text{confidence interval approach}}$$

Case 1: σ^2 is known



Case 1: σ^2 is known

Table 8.1-1 Tests of hypotheses about one mean, variance known

H_0	H_1	Critical Region
$\mu = \mu_0$	$\mu > \mu_0$	$z \geq z_\alpha$ or $\bar{x} \geq \mu_0 + z_\alpha \sigma/\sqrt{n}$
$\mu = \mu_0$	$\mu < \mu_0$	$z \leq -z_\alpha$ or $\bar{x} \leq \mu_0 - z_\alpha \sigma/\sqrt{n}$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\lvert z\rvert \geq z_{\alpha/2}$ or $\lvert\bar{x} - \mu_0\rvert \geq z_{\alpha/2}\sigma/\sqrt{n}$

Case 2: σ^2 is unknown

- Essentially, we replace the standard normal distribution by t-distribution with $d.f. = (n-1)$ and σ^2 by S^2 in Case 1.
- We consider the test statistic

$$T := \frac{\overline{X} - \mu_0}{S/\sqrt{n}}.$$

- Under $H_0 : \mu = \mu_0$, $T \sim t(n-1)$, t distribution with $n-1$ degrees of freedom.
- Once we observe the data $x_1, \ldots, x_n$, we can compute

$$\boxed{t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}}}.$$

Case 2: σ^2 is unknown:

$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$

- p-value is

$$p = P(T > t; H_0) = P(t(n-1) > t),$$

so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = P(t(n-1) > t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \geq t_\alpha(n-1)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin [\overline{x} - t_\alpha(n-1)\frac{s}{\sqrt{n}}, \infty)}}_{\text{confidence interval approach}}$$

Case 2: σ^2 is unknown:

$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$

- p-value is

$$p = P(T < t; H_0) = P(t(n-1) < t),$$

so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = P(t(n-1) < t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \leq -t_\alpha(n-1)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin (-\infty, \overline{x} + t_\alpha(n-1)\frac{s}{\sqrt{n}}]}}_{\text{confidence interval approach}}$$

Case 2: σ^2 is unknown:

$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$

- p-value is

$$\begin{aligned} p &= P(|T| > |t|; H_0) = P(t(n-1) > |t|) + P(t(n-1) < -|t|) \\ &= 2P(t(n-1) > |t|), \end{aligned}$$

so we reject H_0 at significance level α when,

$$\underbrace{\boxed{\text{p-value} = 2P(t(n-1) > |t|) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{|t| \geq t_{\alpha/2}(n-1)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{\mu_0 \notin [\overline{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}, \overline{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}]}}_{\text{confidence interval approach}}$$

Case 2: σ^2 is unknown

Table 8.1-2 Tests of hypotheses for one mean, variance unknown

H_0	H_1	Critical Region
$\mu = \mu_0$	$\mu > \mu_0$	$t \geq t_\alpha(n-1)$ or $\bar{x} \geq \mu_0 + t_\alpha(n-1)s/\sqrt{n}$
$\mu = \mu_0$	$\mu < \mu_0$	$t \leq -t_\alpha(n-1)$ or $\bar{x} \leq \mu_0 - t_\alpha(n-1)s/\sqrt{n}$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$ or $\lvert\bar{x} - \mu_0\rvert \geq t_{\alpha/2}(n-1)s/\sqrt{n}$

Example 8.1-3

Example 3

Example 8.1-3 in book: Let X be the growth of tumor induced on a mouse in 15 days. Suppose that we have $n = 9, \overline{x} = 4.3, s = 1.2$. We would like to test

$$H_0 : \mu = 4, \quad H_1 : \mu \neq 4.$$

We take the significance level $\alpha = 0.1 = 10\%$. Then we have

$$t_{0.05}(8) = 1.86$$

$$t = \frac{4.3 - 4}{1.2/\sqrt{9}} = 0.75$$

Example 8.1-3

- **critical region approach**:

$$|t| = 0.75 < 1.86,$$

and so we fail to reject H_0 at 10% significance level.

- **p-value approach**:

$$p = 2P(t(8) > 0.75) = 0.4747 > 0.1,$$

and so we fail to reject H_0 at 10% significance level.

- **confidence interval approach**: The 90% two-sided C.I. is

$$4.3 \pm 1.86\frac{1.2}{\sqrt{9}} = [3.556, 5.044],$$

which includes 4, and so we fail to reject H_0 at 10% significance level.

Example 8.1-4

Example 4

Example 8.1-4 in book: Suppose that we have $n = 25, \overline{x} = 308.8, s = 115.15$. We would like to test

$$H_0 : \mu = 500, \quad H_1 : \mu < 500.$$

We take the significance level $\alpha = 0.01 = 1\%$. Then we have

$$t_{0.01}(24) = 2.492$$

$$t = \frac{308.8 - 500}{115.15/\sqrt{25}} = -8.30$$

Example 8.1-4

- **critical region approach**:

$$t = -8.30 < -2.492,$$

and so we reject H_0 at 1% significance level.

- **p-value approach**:

$$p = P(t(24) < -8.30) = 8.17 \times 10^{-9} < 0.01,$$

and so we reject H_0 at 1% significance level.

- **confidence interval approach**: The 99% one-sided C.I. is

$$(-\infty, 308.8 + 2.492 \times 115.15/\sqrt{25}] = (-\infty, 366.191],$$

which does not include 500, and so we reject H_0 at 1% significance level.